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Digitizing Zimbabwe's Livestock Trade — a Five-Role Platform for Disease Control, Anti-Theft Verification & More

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Introduction

Livestock trade in Zimbabwe runs on paper, word-of-mouth, and informal trust. January Disease (Theileriosis) alone has killed over 500,000 cattle since 2016. Stock theft is organized enough that a single recent police crackdown produced over 3,400 arrests. Communal farmers — the majority — are routinely excluded from formal credit and verified trade because signup and paperwork assume freehold title they don't hold.

Objective

Build and deploy a role-aware digital platform — Farmer, Veterinarian, Supplier, Retailer, Police — that operationalizes Zimbabwe's actual legal framework for livestock trade (Stock Theft Prevention Act, Animal Health Act, Veterinary Surgeons Act, Consumer Protection Act) as working software, not an afterthought bolted onto a generic farm app.

Methodology

A React (Vite) web app and Expo mobile app share a Flask/MySQL backend with JWT authentication, bcrypt password hashing, and per-owner data scoping. An AI compliance assistant (Jinda) answers legal/health questions from researched Zimbabwean law and refuses to leak other users' data. ESP32/LoRa IoT hardware pairs collars and base stations to a farmer's account for herd health and anti-theft monitoring. The system architecture is shown at right — note the Police Clearance Layer.

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stakeholder roles	backend API endpoints	Zimbabwean law/species docs researched

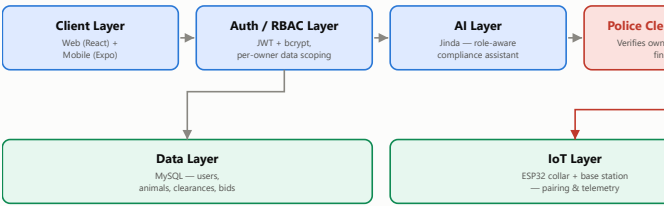
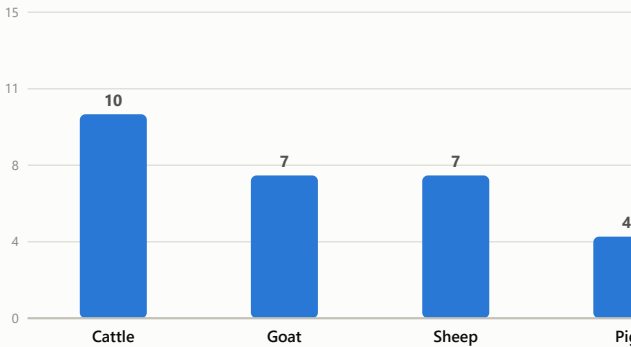


Figure 1. System architecture — the Police Clearance Layer always has final veto over every livestock list. Zimbabwe's real ownership → vet → police clearance sequence.

Disease Coverage by Species

15 notifiable/high-impact diseases modeled in the diagnostics engine, spanning WOAHP-notifiable Zimbabwe-endemic conditions (FMD, January Disease, Anthrax, ASF, PPR, CCPP, and more).



Key Sources & Acknowledgements

Stock Theft Prevention Act [Chapter 9:18], Zimbabwe
Animal Health Act [Chapter 19:01] & Animal Health (General) Regulations, 1994
Veterinary Surgeons Act [Chapter 27:15] — Council of Veterinary Surgeons of Zimbabwe (CVSZ)
Consumer Protection Act [Chapter 14:44], Zimbabwe, 2019

World Organisation for Animal Health
Zimbabwe Republic Police — Anti Stock Theft Unit
Full legal research & source citations: [link]
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